Chapter 7: Introduction to Point Estimation and Testing

Probability and Statistics for Science and Engineering
with Examples in R

Hongshik Ahn

(1) Point Estimation

- (Point) estimator: a statistic intended for estimating a parameter
- (Point) estimate: an observed value of the estimator

Example 7.1:

$\bar{X}$ is an estimator of μ.

$$x_1 = 7.1, x_2 = 5.5, x_3 = 6.6 \Longrightarrow \bar{x} = \frac{7.1 + 5.5 + 6.6}{3} = 6.40 \rightarrow \text{estimate}$$

$$\hat{\sigma}^2 = S^2 = \frac{\sum(X_i - \bar{X})^2}{n-1} = \frac{\sum X_i^2 - (\sum X_i)^2/n}{n-1}\text{: estimator of } \sigma^2$$

$$\frac{\sum x_i^2 - (\sum x_i)^2/n}{n-1} = \frac{\sum x_i^2 - n\bar{x}^2}{n-1} = \frac{(7.1^2 + 5.5^2 + 6.6^2) - 3(6.40)^2}{3-1} = 0.67$$

$\rightarrow$ estimate

- Standard error (s.e.): standard deviation of an estimator

$$E(\bar{X}) = \mu, \quad Var(\bar{X}) = \frac{\sigma^2}{n}$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \Longrightarrow \text{s.e.}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

Unbiased Estimators

An estimator $\hat{\theta}$ is an unbiased estimator of θ if $E(\hat{\theta}) = \theta$ for all θ.
If $\hat{\theta}$ is not unbiased (biased), then $E(\hat{\theta}) - \theta$ is called bias.

Example 7.2: $X \sim \text{Bin}(n, p), \quad n = 25$
Find an unbiased estimator of p.

$$E(X) = np$$

$$E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = \frac{1}{n} \cdot np = p$$

$\therefore \hat{p} = \frac{X}{n}$ is an unbiased estimator of p.

If $x = 15$, then $\frac{x}{n} = \frac{15}{25} = \frac{3}{5}$ is the estimate.

- $X \sim \text{Bin}(n, p) \Longrightarrow \hat{p} = \frac{X}{n}$ is an unbiased estimator of p.

The s.e. of $\hat{p} = \frac{X}{n}$:

$$\sigma_{\hat{p}} = \sqrt{Var\left(\frac{X}{n}\right)} = \sqrt{\frac{Var(X)}{n^2}} = \sqrt{\frac{np(1-p)}{n^2}} = \sqrt{p(1-p)/n}$$

s.e. estimator: $\hat{\sigma}_{\hat{p}} = \sqrt{\hat{p}(1-\hat{p})/n}$

s.e. estimate: $\sqrt{(0.6)(0.4)/25} = 0.098$

<u>Example 7.3:</u> In a presidential election poll conducted the day before the election, a sample of 1500 voters were asked for their intended vote. From this sample, 45% preferred the candidate from a conservative party. Assume everyone in the sample expressed a preference.

The unbiased estimate of this ratio is 0.45, and the estimate of s.e. is $\sqrt{(0.45)(0.55)/1500} = 0.013$.

- Let $X_1, \cdots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. Then

  $\hat{\mu} = \bar{X}$ is an unbiased estimator of μ and

  $\hat{\sigma}^2 = S^2 = \frac{\sum(X_i - \bar{X})^2}{n-1}$ is an unbiased estimator of σ^2.

(2) Tests of Hypotheses

A statistical hypothesis is a statement about a population parameter.

(A) The null and the alternative hypotheses

Example 7.4: The cure rate for a given disease using a standard medication is 35%. The cure rate of a new drug is claimed to be better. A sample has $n = 18$ patients.
X: number of patients who were cured
Is there substantial evidence that the new drug has a higher cure rate than the standard medication?

H_0: The new drug is not better: $p \leq 0.35$ (null hypothesis)
H_1: The new drug is better: $p > 0.35$ (alternative hypothesis)

Choice of H_0 & H_1

When our goal is to establish an assertion, the <u>negation</u> of the assertion is H_0, and the <u>assertion</u> itself is H_1.

If H_0 is true, we would expect 6 or fewer cures out of 18.
If H_1 is true, we would expect more than 6 cures out of 18.

<u>Decision rule</u> for a test of the null hypothesis:

Reject H_0 if $X > c$

Do not reject H_0 if $X \leq c$

(X: test statistic; $\{X > c\}$: Rejection region)

- <u>Test</u>: decision rule that tells us when to reject H_0, when not to reject H_0
- <u>Test statistic</u>: a statistic whose value serves to determine the action
- <u>Rejection region</u>: the set of values of a test statistic which H_0 is to be rejected

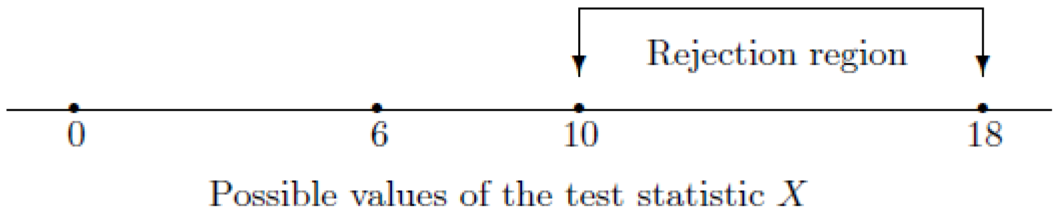
(B) The Two Types of Errors

	Unknown True State of Nature	
Test Concludes	H_0 is True	H_0 is False
Do not Reject H_0	Correct	Wrong (Type II error)
Reject H_0	Wrong (Type I error)	Correct

Type I error: reject H_0 when H_0 is true

Type II error: fail to reject H_0 when H_1 is true

Example 7.4 (continued): $H_0: p \leq 0.35$ vs. $H_1: p > 0.35$, $n = 18$
Rejection region: $R: X \geq 10$
Determine the type of error that can occur and calculate the error probability when (a) $p = 0.3$, (b) $p = 0.6$.



(a) $p = 0.3 \Longrightarrow H_0$ is true. Type I error: Reject H_0

$P(\text{type I error given } p = 0.3) = P(X \geq 10 \text{ given } p = 0.3)$

$X \sim \text{Bin}(18, 0.3)$

$P(X \geq 10) = 1 - P(X \leq 9) = 1 - 0.979 = 0.021$

$P(\text{type I error given } p = 0.3) = 0.021$

(b) $p = 0.6 \Longrightarrow H_1$ is true. Type II error: Accept H_0

$X \sim \text{Bin}(18, 0.6)$

$P(\text{type II error given } p = 0.6) = P(X \leq 9 \text{ given } p = 0.6) = 0.263$

Example 7.4 (Continued):

	p in H_0		
Type I error Probability	0.2	0.3	0.35
$\boldsymbol{P(X \geq 10)}$	0.001	0.021	0.060

Level of significance (significance level; α):
The max type I error probability of a test

In the above example, $\alpha = 0.060.$

Type II error probability: $\beta = P$(type II error)

Power: $1 - \beta$

Example 7.5: The lifetime of a certain type of car battery: $N(5, 0.9^2)$ (yr.)
A new kind of battery is designed to increase the average lifetime.

$$H_0: \mu = 5, H_1: \mu > 5$$

$X_1, \cdots, X_{25}$ $\quad \bar{X} \sim N\left(\mu, \left(\sigma/\sqrt{n}\right)^2\right) = N(\mu, 0.18^2)$

Suppose rejection region: $\{\bar{X} \geq 5.42\}$. Calculate the type I error rate and also the type II error rates when $\mu = 5.2$ or 5.5.

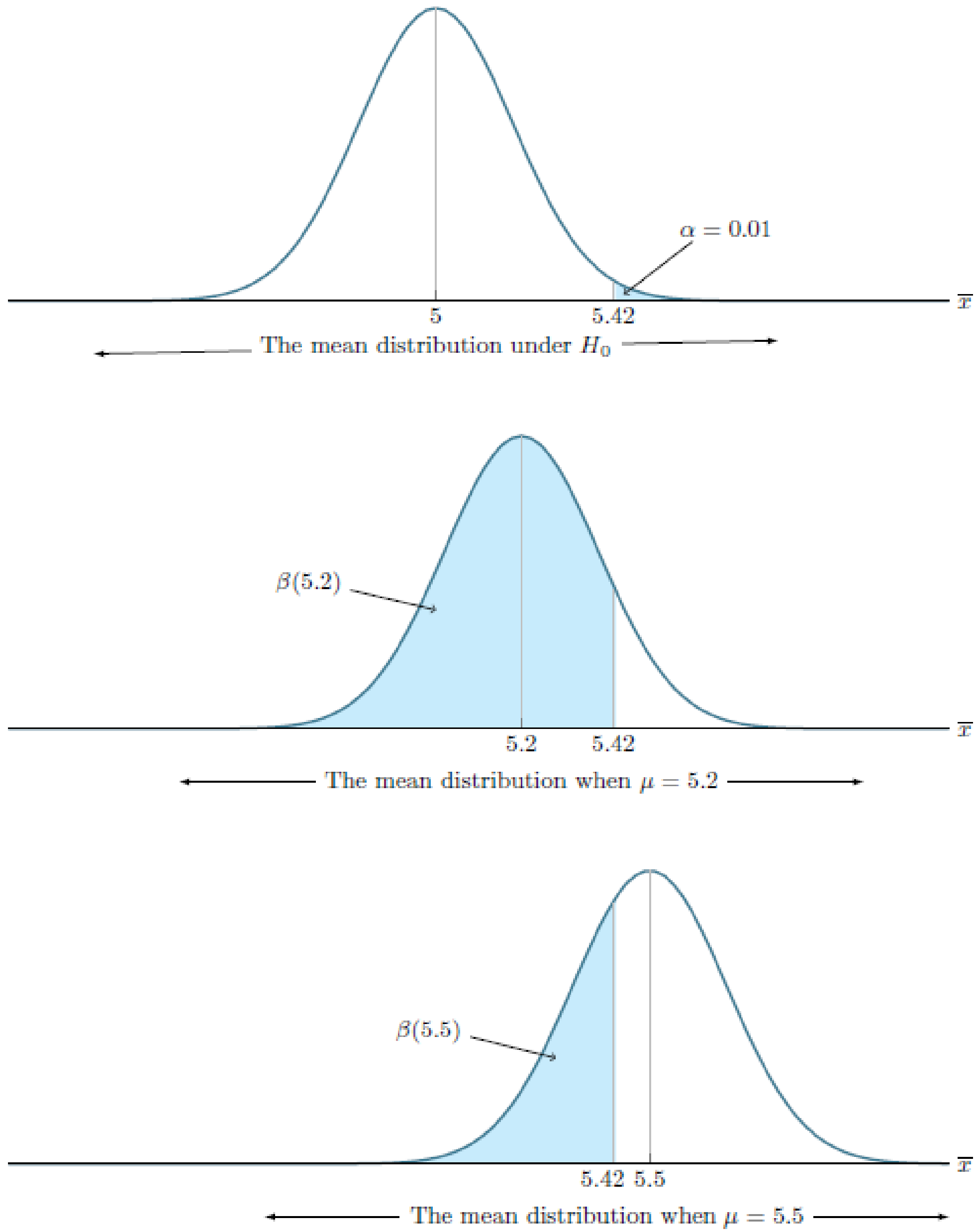
$\alpha = P$(type I error) $= P(H_0$ is rejected when it is true)
$= P(\bar{X} \geq 5.42$ when $\bar{X} \sim N(5, 0.18^2))$
$= P\left(Z \geq \frac{5.42-5}{0.18}\right) = P(Z \geq 2.33) = \Phi(-2.33) = 0.01$

$\beta(5.2) = P(\text{Type II error when } \mu = 5.2)$
$= P(\text{do not reject } H_0$ when $\mu = 5.2)$
$= P(\bar{X} < 5.42 \text{ when } \bar{X} \sim N(5.2, 0.18^2))$
$= P\left(Z < \frac{5.42-5.2}{0.18}\right) = \Phi(1.22) = 0.8888$

Power: $1 - 0.8888 = 0.1112$

$\beta(5.5) = P\left(Z < \frac{5.42-5.5}{0.18}\right) = \Phi(-0.44) = 0.33$

Power: $1 - 0.33 = 0.67$



$\alpha = 0.01$
5
5.42
$\overline{x}$
The mean distribution under H_0
$\beta(5.2)$
5.2
5.42
$\overline{x}$
The mean distribution when $\mu = 5.2$
$\beta(5.5)$
5.42
5.5
$\overline{x}$
The mean distribution when $\mu = 5.5$

Suppose an experiment and n are fixed. Then decreasing the size of the rejection region $\Longrightarrow$ smaller $\alpha \Longrightarrow$ larger β

Testing Hypothesis (overall), 5 Steps:

<u>Step 1</u> Setup H_0 & H_1 ←question

<u>Step 2</u> α = ?

<u>Step 3</u> Choose the test statistic and critical (rejection) region

<u>Step 4</u> Substitute the values in the test statistic

Decision

<u>Step 5</u> Calculate the p-value

- The p-value is the probability of obtaining a value for the test statistic that is more extreme than the value actually observed. (Probability is calculated under the null hypothesis H_0.)

- Meaning of the p-value:

  Reject H_0 if $p \leq \alpha$

  Do not reject H_0 if $p > \alpha$